

put $x_3 = 2$ in (2)

$$x_2 - 3x_3 = -3$$

$$x_2 - 3(2) = -3$$

$$\boxed{x_2 = 3}$$

put x_1 and x_2 in (1)

$$x_1 + x_2 + 2x_3 = 1$$

$$x_1 + 3 + 2(2) = 1$$

$$x_1 + 7 = 1$$

$$x_1 = -6$$

As system is consistent and yield unique solution so u_1, u_2 and u_3 are Linear combi of V

$$V = -6u_1 + 3u_2 + 2u_3$$

Example:

Express $v = (2, -5, 3)$ in $\mathbb{R}^3$ as LC of of vectors

$$u_1 = (1, 3, 2), u_2 = (2, -4, -1), u_3 = (1, -5, 7)$$

$$au_1 + bu_2 + cu_3 = V$$

$$\left[\begin{array}{ccc|c} 1 & 2 & 1 & 2 \\ -3 & -4 & -5 & -5 \\ 2 & -1 & 7 & 3 \end{array} \right] \rightarrow \left[\begin{array}{ccc|c} 1 & 2 & 1 & 2 \\ 0 & 2 & -2 & 1 \\ 0 & -5 & 5 & -1 \end{array} \right] \begin{array}{l} R_2 + 3R_1 \\ R_3 - 2R_1 \end{array}$$

$$\left[\begin{array}{ccc|c} 1 & 2 & 1 & 2 \\ 0 & 1 & -1 & \frac{1}{2} \\ 0 & -5 & 5 & -1 \end{array} \right] \xrightarrow[R_2]{R_2 \times 2} \left[\begin{array}{ccc|c} 1 & 2 & 1 & 2 \\ 0 & 1 & -1 & \frac{1}{2} \\ 0 & 0 & 0 & \frac{3}{2} \end{array} \right] \begin{array}{l} \\ R_3 + 5R_2 \end{array}$$

since $a \neq -\frac{3}{2}$ so system is inconsistent
 & we can't write linear combination.

4.6 Express polynomial $v = t^2 + 4t - 3$ in
 $P(t)$ as a linear combination of polynomial

$$P_1 = t^2 - 2t + 5, P_2 = 2t^2 - 3t, P_3 = t + 1$$

set v as a LC of P_1, P_2, P_3 using unknown
 x, y, z to obtain

$$t^2 + 4t - 3 = x(t^2 - 2t + 5) + y(2t^2 - 3t) + z(t + 1)$$

$$t^2 + 4t - 3 = xt^2 - 2xt + 5x + 2yt^2 - 3yt + zt + z$$

$$t^2 + 4t - 3 = (x + 2y)t^2 + (-2x - 3y + z)t + (5x + z)$$

compare:

$$x + 2y = 1$$

$$-2x - 3y + z = 4$$

$$5x + z = -3$$

$$\left[\begin{array}{ccc|c} 1 & 2 & 0 & 1 \\ -2 & -3 & 1 & 4 \\ 5 & 0 & 1 & -3 \end{array} \right] \rightarrow \left[\begin{array}{ccc|c} 1 & 2 & 0 & 1 \\ 0 & 1 & 1 & 6 \\ 0 & -10 & 1 & -8 \end{array} \right] \begin{array}{l} \\ R_2 + 2R_1 \\ R_3 - 5R_1 \end{array}$$

$$\left[\begin{array}{ccc|c} 1 & 2 & 0 & 1 \\ 0 & 1 & 1 & 6 \\ 0 & 0 & 11 & 52 \end{array} \right] \begin{array}{l} \\ R_3 + 10R_2 \\ \end{array}$$

$$\left[\begin{array}{ccc|c} 1 & 2 & 0 & 1 \\ 0 & 1 & 1 & 6 \\ 0 & 0 & 1 & \frac{52}{11} \end{array} \right]$$

$$x + 2y = 1$$

$$y + z = 6$$

$$z = \frac{52}{11}$$

$$x + 2y = 1$$

$$x + 2\left(\frac{14}{11}\right) = 1$$

$$x = \frac{-28}{11} + 1$$

$$x = \frac{-28 + 11}{11}$$

$$x = \frac{-17}{11}$$

$$y + z = 6$$

$$y + \frac{52}{11} = 6$$

$$y = \frac{66 - 52}{11}$$

$$y = \frac{14}{11}$$

So system is consistent and yields unique solution Hence v is a linear combination P_1, P_2, P_3 .

$$V = \frac{-17}{11}P_1 + \frac{14}{11}P_2 + \frac{52}{11}P_3$$

4.7- Express M as a LC of matrices A, B, C where

$$M = \begin{bmatrix} 4 & 7 \\ 7 & 9 \end{bmatrix} \text{ and } A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}, B = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

$$C = \begin{bmatrix} 1 & 1 \\ 4 & 5 \end{bmatrix}$$

$$xA + yB + zC = M$$

$$x \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} + y \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} + z \begin{bmatrix} 1 & 1 \\ 4 & 5 \end{bmatrix} = \begin{bmatrix} 4 & 7 \\ 7 & 9 \end{bmatrix}$$

$$\begin{bmatrix} x+y+z & x+2y+z \\ x+3y+4z & x+4y+5z \end{bmatrix} = \begin{bmatrix} 4 & 7 \\ 7 & 9 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 & 1 & 4 \\ 1 & 2 & 1 & 7 \\ 1 & 3 & 4 & 7 \\ 1 & 4 & 5 & 9 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 & 1 & 4 \\ 0 & 1 & 0 & 3 \\ 0 & 2 & 3 & 3 \\ 0 & 3 & 4 & 5 \end{bmatrix} \begin{array}{l} R_2 - R_1 \\ R_3 - R_1 \\ R_4 - R_1 \end{array}$$

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$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 4 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 3 & -3 \\ 0 & 0 & 4 & -4 \end{array} \right] \begin{array}{l} \\ \\ R_3 - 2R_2 \\ R_4 - 3R_2 \end{array}$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 4 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 4 & -4 \end{array} \right] \begin{array}{l} \\ R_3 \times 3 \\ \\ \end{array}$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 4 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 \end{array} \right] \begin{array}{l} \\ R_4 - 4R_1 \\ \\ \end{array}$$

Leading Var = x, y, z

$$x + y + z = 4$$

$$y = 3$$

$$z = -1$$

$$x + y + z = 4$$

$$x + 3 - 1 = 2$$

$$x + 2 = 4$$

$$x = 2$$

so

$$M = 2A + 3B - C$$

Linear Independence

4.17. Determine whether or not u & v are linearly dependent where:
Two vectors are dependent iff one is scalar multiple of other.

a-

$$u = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, v = \begin{bmatrix} 3 \\ -5 \end{bmatrix}$$

Since any of one is doesn't depend on other so Linearly independent.

b-

$$u = \begin{bmatrix} 1 \\ -3 \end{bmatrix}, v = \begin{bmatrix} -2 \\ 6 \end{bmatrix}$$

$$v = -2u$$

so Linearly Dependent

c-

$$u = \begin{bmatrix} 1 \\ 2 \\ -3 \end{bmatrix}, v = \begin{bmatrix} 4 \\ 5 \\ -6 \end{bmatrix}$$

Linearly Independent.

4-20 - Determine List of vector $\mathbb{R}^3$ is Linearly Dependent.

a-

$$u_1 = (1, 2, 5), u_2 = (1, 3, 1), u_3 = (2, 5, 7) \\ u_4 = (3, 1, 4)$$

-b

→ Any set of ~~Matrix~~ more than vectors in $\mathbb{R}^3$ is Linearly Dependent coz free variables get involved and there will be no x, y, z becomes 0.

b-

-P1-1

$$u = (1, 2, 5), v = (2, 5, 1) \text{ and } w = (1, 5, 2)$$

$$xu + yv + zw = 0$$

when you solve you get $x, y, z = 0$ so Linearly independent.

c-

$$u = (1, 2, 3), v = (0, 0, 0), w = (1, 5, 6)$$

Because 0 is one of vectors so Linearly Dependent coz Free var get involved.

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Example: 3 (Linear Independence $\mathbb{R}^4$)

$$v_1 = \begin{bmatrix} 1 \\ 2 \\ 2 \\ -1 \end{bmatrix}, v_2 = \begin{bmatrix} 4 \\ 9 \\ 9 \\ -4 \end{bmatrix}, v_3 = \begin{bmatrix} 5 \\ 8 \\ 9 \\ -5 \end{bmatrix}$$

$$xv_1 + yv_2 + zv_3 = 0$$

$$\left[\begin{array}{ccc|c} 1 & 4 & 5 & 0 \\ 2 & 9 & 8 & 0 \\ 2 & 9 & 9 & 0 \\ -1 & -4 & -5 & 0 \end{array} \right]$$

$$\left[\begin{array}{ccc|c} 1 & 4 & 5 & 0 \\ 0 & 1 & -2 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right] \begin{array}{l} R_2 - 2R_1 \\ R_3 - 2R_1 \\ R_4 + R_1 \end{array}$$

$$\left[\begin{array}{ccc|c} 1 & 4 & 5 & 0 \\ 0 & 1 & -2 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right] \begin{array}{l} \\ R_3 - R_2 \\ \\ \end{array}$$

Leading var = x, y, z

$$x + 4y + 5z = 0$$

$$y - 2z = 0$$

$$z = 0$$

$$y - 2z = 0$$

$$x + 4y + 5z = 0$$

$$y - 2(0) = 0 \quad x + 4(0) + 5(0) = 0$$

$$y = 0$$

$$x = 0$$

Since $x, y, z = 0$ so Linearly
Independent

Example: 5 (Linear Independence of Polynomials)

Determine whether

$$P_1 = 1-x, P_2 = 5+3x-2x^2, P_3 = 1+3x-x^2$$

are Linearly Independent or Dependent

$$P_1 + P_2 + P_3 = 0$$

$$a(1-x) + b(5+3x-2x^2) + c(1+3x-x^2) = 0$$

$$a - ax + 5b + 3bx - 2bx^2 + c + 3cx - cx^2 = 0$$

$$(a+5b+c) + (-a+3b+3c)x + (-2b-c)x^2 = 0$$

$$a + 5b + c = 0$$

$$-a + 3b + 3c = 0$$

$$-2b - c = 0$$

$$\left[\begin{array}{ccc|c} 1 & 5 & 1 & 0 \\ -1 & 3 & 3 & 0 \\ 0 & -2 & -1 & 0 \end{array} \right] \rightarrow \left[\begin{array}{ccc|c} 1 & 5 & 1 & 0 \\ 0 & 8 & 4 & 0 \\ 0 & -2 & -1 & 0 \end{array} \right] R_2 + R_1$$

$$\left[\begin{array}{ccc|c} 1 & 5 & 1 & 0 \\ 0 & 1 & \frac{1}{2} & 0 \\ 0 & -2 & -1 & 0 \end{array} \right] R_2 \rightarrow \left[\begin{array}{ccc|c} 1 & 5 & 1 & 0 \\ 0 & 1 & \frac{1}{2} & 0 \\ 0 & 0 & 0 & 0 \end{array} \right] R_3 + 2R_2$$

Leading Var = a, b

Free Var = c

Let

$$c = t$$

$$a + 5b + c = 0$$

$$a + 5\left(-\frac{1}{2}t\right) + t = 0$$

$$a = 5\frac{t}{2} - t = 0$$

$$a = 3\frac{t}{2}$$

$$b + \frac{1}{2}c = 0$$

$$b + \frac{1}{2}t = 0$$

$$b = -\frac{1}{2}t$$

$t \in \mathbb{R}$, so Linearly dependent

Basis & Dimension

If $S = \{v_1, v_2, v_3, \dots, v_n\}$ is set of vectors in finite dimensional vectors space V the S is called a basis for V if

(a) S spans V (Linear Combination)

(b) S is Linearly Independent.

When S spans V ?

A set S spans vector space V if every vector in V can be written as Linear combination of vectors in S .

When Form Basis.

→ S spans V

→ S is Linear Independence.

Example 4: (Standard Basis)

show matrices are form Basis

$$M_1 = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, M_2 = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, M_3 = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$$

$$M_4 = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$

Check Linear Independence

$$xM_1 + yM_2 + zM_3 + wM_4 = 0$$

$$x \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + y \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} + z \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} + w \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} x+0+0+0 & 0+y+0+0 \\ 0+0+z+0 & 0+0+0+w \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$\begin{array}{l} x=0 \\ y=0 \\ z=0 \\ w=0 \end{array} \longleftrightarrow \begin{bmatrix} 1 & 0 & 0 & 0 & | & 0 \\ 0 & 1 & 0 & 0 & | & 0 \\ 0 & 0 & 1 & 0 & | & 0 \\ 0 & 0 & 0 & 1 & | & 0 \end{bmatrix}$$

as scalars are already 0 so ~~these~~ matrices
Linearly Independent.

$$\text{Rank} = 4$$

Check Linear Combination

$$xM_1 + yM_2 + zM_3 + wM_4 = M$$

$$x \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + y \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} + z \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} + w \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 & a \\ 0 & 1 & 0 & 0 & b \\ 0 & 0 & 1 & 0 & c \\ 0 & 0 & 0 & 1 & d \end{bmatrix} \quad \begin{matrix} M_{22} = aM_1 + bM_2 \\ \quad \quad \quad cM_3 + dM_4 \end{matrix}$$

AS consistent so we can write
Linear Combination so Matrices spans M_{22} .

Thus matrices Form Basis as

- Matrices are Linearly Independent
- Matrices Spans M_{22}

4.24-

Determine whether or not each of following form Basis of $\mathbb{R}^3$.

Rule:- The number of vector in elements in any Basis of vector space must equal to the Dimension of Space.

For $\mathbb{R}^3$

must have 3 vector not more

not less

a-

$$(1, 1, 1), (1, 0, 1)$$

Not form basis - coz basis of $\mathbb{R}^3$ must contain exactly 3 elements

b-

No, same reason as "a"

c-

$$v_1 = (1, 1, 1), v_2 = (1, 2, 3), v_3 = (2, -1, 1)$$

$$\lambda v_1 + \lambda v_2 + \lambda v_3 = 0$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 2 & 0 \\ 1 & 2 & -1 & 0 \\ 1 & 3 & 1 & 0 \end{array} \right] \rightarrow \left[\begin{array}{ccc|c} 1 & 1 & 2 & 0 \\ 0 & 1 & -3 & 0 \\ 0 & 2 & -1 & 0 \end{array} \right] \begin{array}{l} R_2 - R_1 \\ R_3 - R_1 \end{array}$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 2 & 0 \\ 0 & 1 & -3 & 0 \\ 0 & 0 & 5 & 0 \end{array} \right] \xrightarrow{R_3 - 2R_2} \left[\begin{array}{ccc|c} 1 & 1 & 2 & 0 \\ 0 & 1 & -3 & 0 \\ 0 & 0 & 1 & 0 \end{array} \right] \begin{array}{l} R_3 / 5 \end{array}$$

$$x + y + 2z = 0$$

$$y - 3z = 0$$

$$z = 0$$

$$y - 3z = 0$$

$$y = 0$$

$$x + y + 2z = 0$$

$$x + 0 + 2(0) = 0$$

$$x = 0$$

all are 0 so linearly independent

Linear combination.

$$xv_1 + yv_2 + zv_3 = M$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 2 & a \\ 1 & 2 & -1 & b \\ 1 & 3 & 1 & c \end{array} \right] \rightarrow \left[\begin{array}{ccc|c} 1 & 1 & 2 & a \\ 0 & 1 & -3 & b-a \\ 0 & 2 & -1 & c-a \end{array} \right] \begin{array}{l} R_2 - R_1 \\ R_3 - R_1 \end{array}$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 2 & a \\ 0 & 1 & -3 & b-a \\ 0 & 0 & 5 & c+a-2b \end{array} \right] R_3 - 2R_2$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 2 & a \\ 0 & 1 & -3 & b-a \\ 0 & 0 & 1 & \frac{c+a-2b}{5} \end{array} \right] R_3 \times 5$$

so consistent Hence vectors span M.

so vectors form basis

d-

same as part e

4-25-

$$(1, 1, 1), (1, 2, 3, 2), (2, 5, 6, 4) \\ (2, 6, 8, 5)$$

same as part d. and e of previous qg.

Example: B

$$B = \left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix} \right\}$$

Basis For $\mathbb{R}^3$

1. check Linear Dependency

$$xv_1 + yv_2 + zv_3 = 0$$

2. check Linear Combination

$$xv_1 + yv_2 + zv_3 = w$$

Example: 4

$$S = \left\{ \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} \right\}$$

is Base For W .

*note: Trace is 0 given

$$xM_1 + yM_2 + zM_3 = 0$$

$$x \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} + y \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} + z \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} x & y \\ z & -x \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ -1 & 0 & 0 & 0 \end{array} \right] \rightarrow \left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

$x \neq y, z=0$ so Independent

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$$xM_1 + yM_2 + zM_3 = W$$

is a linear combination

$$\begin{bmatrix} x & y \\ z & -x \end{bmatrix} = \begin{bmatrix} a & b \\ c & -a \end{bmatrix}$$

$$\begin{array}{ccc|c} 1 & 0 & 0 & a \\ 0 & 1 & 0 & b \\ 0 & 0 & 1 & c \\ -\phi & 0 & 0 & -a \end{array} \rightarrow \begin{array}{ccc|c} 1 & 0 & 0 & a \\ 0 & 1 & 0 & b \\ 0 & 0 & 1 & c \\ 0 & 0 & 0 & 0 \end{array} \quad R_4 + R_1$$

consistent

So

form basis and span S V

Example: 5

$$B = \left\{ \begin{bmatrix} 1 & 3 \\ 2 & 1 \end{bmatrix}, \begin{bmatrix} -1 & 2 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & -4 \end{bmatrix} \right\}$$

basis for $M_{2 \times 2}$.

$$\left[\begin{array}{ccc|c} 1 & -1 & 0 & 0 \\ 3 & 2 & 1 & 0 \\ 2 & 1 & 0 & 0 \\ 1 & 0 & -4 & 0 \end{array} \right] \rightarrow \left[\begin{array}{ccc|c} 1 & -1 & 0 & 0 \\ 0 & 5 & 1 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 1 & -4 & 0 \end{array} \right] \begin{array}{l} R_2 - 3R_1 \\ R_3 - 2R_1 \\ R_4 - R_1 \end{array}$$

$$\left[\begin{array}{ccc|c} 1 & -1 & 0 & 0 \\ 0 & 1 & -4 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 5 & 1 & 0 \end{array} \right] \begin{array}{l} \\ R_2 \leftrightarrow R_4 \\ \\ \end{array}$$

$$\left[\begin{array}{ccc|c} 1 & -1 & 0 & 0 \\ 0 & 1 & -4 & 0 \\ 0 & 0 & 12 & 0 \\ 0 & 0 & 21 & 0 \end{array} \right] \begin{array}{l} \\ \\ R_3 - 3R_2 \\ R_4 - 5R_2 \end{array}$$

$$\left[\begin{array}{ccc|c} 1 & -1 & 0 & 0 \\ 0 & 1 & -4 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 21 & 0 \end{array} \right] \begin{array}{l} \\ \\ R_3 \times 12 \\ \\ \end{array}$$

$$\left[\begin{array}{ccc|c} 1 & -1 & 0 & 0 \\ 0 & 1 & -4 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right] \begin{array}{l} \\ R_4 - 21R_3 \\ \\ \end{array}$$

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$$x_1 - x_2 = 0$$

$$x_2 - 4x_3 = 0$$

$$x_3 = 0$$

$$x_2 - 4x_3 = 0$$

$$x_2 = 0$$

$$x_1 - x_2 = 0$$

$$x_1 = 0$$

$x_1, x_2, x_3 = 0$ so Linearly Independent

$$x_1 M_1 + x_2 M_2 + x_3 M_3 = M_{22}$$

$$\begin{bmatrix} 1 & -1 & 0 & a \\ 3 & 2 & 1 & b \\ 2 & 1 & 0 & c \\ -1 & 0 & -4 & d \end{bmatrix}$$

$$\begin{bmatrix} 1 & -1 & 0 & a \\ 0 & 5 & 1 & b-3a \\ 0 & 3 & 0 & c-2a \\ 0 & 1 & -4 & d-a \end{bmatrix} \begin{array}{l} R_2 - 3R_1 \\ R_3 - 2R_1 \\ R_4 - R_1 \end{array}$$

$$\begin{bmatrix} 1 & -1 & 0 & a \\ 0 & 1 & -4 & d-a \\ 0 & 3 & 0 & c-2a \\ 0 & 5 & 1 & b-3a \end{bmatrix} \begin{array}{l} \\ R_2 \leftrightarrow R_4 \\ \end{array}$$

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$$\left[\begin{array}{cccc|c} 1 & -1 & 0 & a & \\ 0 & 1 & -4 & d-a & \\ 0 & 0 & 1 & c-2a-3(d-a) & R_3-3R_2 \\ 0 & 0 & 2 & b-3a-5(d-a) & R_4-5R_2 \end{array} \right]$$

$$\left[\begin{array}{cccc|c} 1 & -1 & 0 & a & \\ 0 & 1 & -4 & d-a & \\ 0 & 0 & 1 & (c+2a-3d)/12 & R_3/12 \\ 0 & 0 & 2 & (b+2a-5d)/12 & R_4/12 \end{array} \right]$$

$$\left[\begin{array}{cccc|c} 1 & -1 & 0 & a & \\ 0 & 1 & -4 & d-a & \\ 0 & 0 & 1 & (c+2a-3d)/12 & \\ 0 & 0 & 0 & \text{Alger Express} & R_4-2R_3 \end{array} \right]$$

so $0 \neq$ Algebraic inconsistent
does not ~~span~~ form basis for $\mathbb{R}^4$.

Example : 3 (Another Basis for $\mathbb{R}^3$)

Show

$$v_1 = (1, 2, 1), v_2 = (2, 9, 0)$$

$$v_3 = (3, 3, 4) \text{ form Basis.}$$

$$xv_1 + yv_2 + zv_3 = 0$$

$$\left[\begin{array}{ccc|c} 1 & 2 & 3 & 0 \\ 2 & 9 & 3 & 0 \\ 1 & 0 & 4 & 0 \end{array} \right]$$

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$$\left[\begin{array}{ccc|c} 1 & 2 & 3 & 0 \\ 0 & 5 & -3 & 0 \\ 0 & -2 & 1 & 0 \end{array} \right] \begin{array}{l} \\ R_2 - 2R_1 \\ R_3 - R_1 \end{array}$$

$$\left[\begin{array}{ccc|c} 1 & 2 & 3 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & -2 & 1 & 0 \end{array} \right] \begin{array}{l} \\ R_2 + 2R_3 \\ \end{array}$$

$$\left[\begin{array}{ccc|c} 1 & 2 & 3 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & -1 & 0 \end{array} \right] \begin{array}{l} \\ R_3 + 2R_2 \\ \end{array}$$

$$\left[\begin{array}{ccc|c} 1 & 2 & 3 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & 0 \end{array} \right] \begin{array}{l} \\ -R_3 \\ \end{array}$$

$$x + 2y + 3z = 0$$

$$y - z = 0$$

$$z = 0$$

$$y - z = 0$$

$$y = 0$$

$$x + 2y + 3z = 0$$

$$x = 0$$

$x, y, z = 0$ So

Independent

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$$2V_1 + 4V_2 + 2V_3 = W$$

$$\left[\begin{array}{ccc|c} 1 & 2 & 3 & a \\ 2 & 9 & 3 & b \\ 1 & 0 & 4 & c \end{array} \right] \rightarrow \left[\begin{array}{ccc|c} 1 & 2 & 3 & a \\ 0 & 5 & -3 & b-2a \\ 0 & -2 & 1 & c-a \end{array} \right] \begin{array}{l} R_2 - 2R_1 \\ R_3 - R_1 \\ G - a \end{array}$$

$$\left[\begin{array}{ccc|c} 1 & 2 & 3 & a \\ 0 & 1 & -1 & b-4a+2c \\ 0 & -2 & 1 & c-a \end{array} \right] \begin{array}{l} R_2 + 2R_3 \\ b-4a+2c \\ 1- \end{array}$$

$$\left[\begin{array}{ccc|c} 1 & 2 & 3 & a \\ 0 & 1 & -1 & b-4a+2c \\ 0 & 0 & -1 & (c-a)+2(b-4a+2c) \end{array} \right] \begin{array}{l} \\ R_2 + 2R_3 \\ \end{array}$$

$$\left[\begin{array}{ccc|c} 1 & 2 & 3 & a \\ 0 & 1 & -1 & b-4a+2c \\ 0 & 0 & -1 & -9a+2b+5c \end{array} \right]$$

$$\left[\begin{array}{ccc|c} 1 & 2 & 3 & a \\ 0 & 1 & -1 & b-4a+2c \\ 0 & 0 & 1 & 9a-2b-5c \end{array} \right] \begin{array}{l} \\ \\ -R_3 \end{array}$$

consistent

So vector span $\mathbb{R}^3$

Example: 7

$$B = \left\{ \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}$$

same as previous questions.

Example: 8

Let

$$S = \left\{ \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} -1 \\ 1 \\ -2 \end{bmatrix} \right\}$$

Find basis for span of S.

$$\begin{bmatrix} 1 & 0 & 1 & 1 & -1 \\ 0 & 1 & 1 & 2 & 1 \\ 1 & 1 & 2 & 1 & -2 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 1 & 1 & -1 \\ 0 & 1 & 1 & 2 & 1 \\ 0 & 1 & 1 & 0 & -1 \end{bmatrix} \quad R_3 - R_1$$

$$\begin{bmatrix} 1 & 0 & 1 & 1 & -1 \\ 0 & 1 & 1 & 2 & 1 \\ 0 & 0 & 0 & -2 & -2 \end{bmatrix} \quad R_3 - R_2$$

NOTES

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$\rightarrow C_1$ has pivot (1)
 C_2 has pivot (1)
 C_4 has pivot (1)

$$\left[\begin{array}{ccccc|c} 1 & 0 & 1 & 1 & -1 & \\ 0 & 1 & 1 & 2 & 1 & \\ 0 & 0 & 0 & 1 & 1 & \end{array} \right] \begin{array}{l} \\ -R_3 \times \frac{1}{2} \\ \end{array}$$

The pivot is on 1st, 2nd & 4th col. so 1st, 2nd and 4th original vectors from set S form basis.

$$B = \left\{ \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} \right\}$$

Example: 9

Find Basis for $\mathbb{R}^4$ that contains vectors.

$$v_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix}$$

$$v_2 = \begin{bmatrix} -1 \\ 1 \\ -1 \\ 0 \end{bmatrix}$$

$$e_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$e_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}$$

Use: e_1, e_2, e_3, e_4

$$e_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}$$

$$\left[\begin{array}{cccccc|c} 1 & -1 & 1 & 0 & 0 & 0 & \\ 0 & 1 & 0 & 1 & 0 & 0 & \\ 1 & -1 & 0 & 0 & 1 & 0 & \\ 0 & 0 & 0 & 0 & 0 & 1 & \end{array} \right]$$

$$e_4 = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

$$\left[\begin{array}{cccccc|c} 1 & -1 & 1 & 0 & 0 & 0 & \\ 0 & 1 & 0 & 1 & 0 & 0 & \\ 0 & 0 & -1 & 0 & 1 & 0 & \\ 0 & 0 & 0 & 0 & 0 & 1 & \end{array} \right] \begin{array}{l} \\ R_3 - R_1 \\ \end{array}$$

Q. 1 (a) (c)

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$$\begin{bmatrix} 1 & -1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \xrightarrow{-R_3}$$

c_1 has pivot

c_2 has pivot

c_3 has pivot

c_6 has pivot

The pivot is on c_1, c_2, c_3 and c_4 so
1 and 2nd from original vectors and
 e_1 and e_4 form basis

$$B = \left\{ \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 1 \\ -1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} \right\}$$